Pose Estimation during RLV Launch and Horizontal-Landing using Camera Arrays

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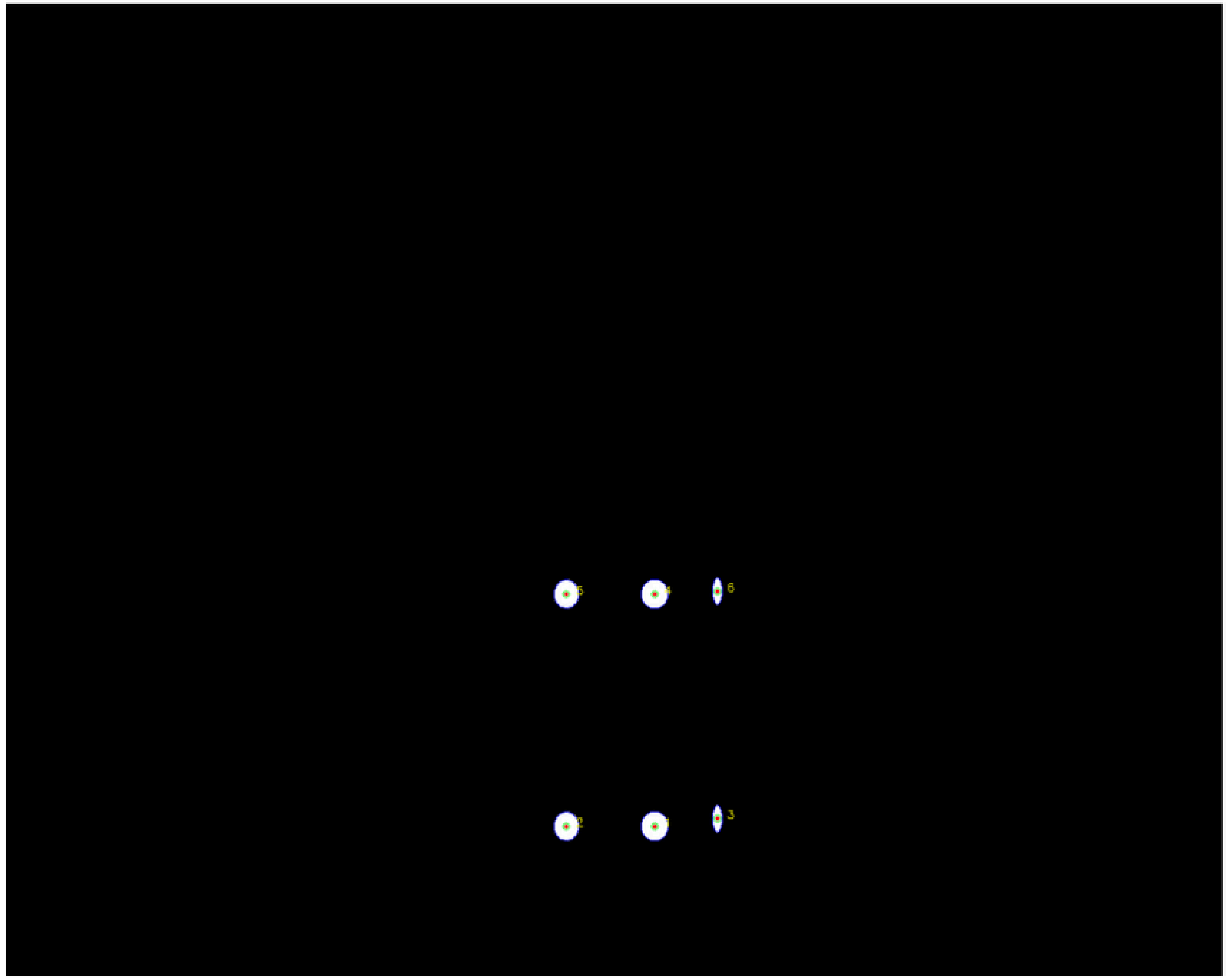
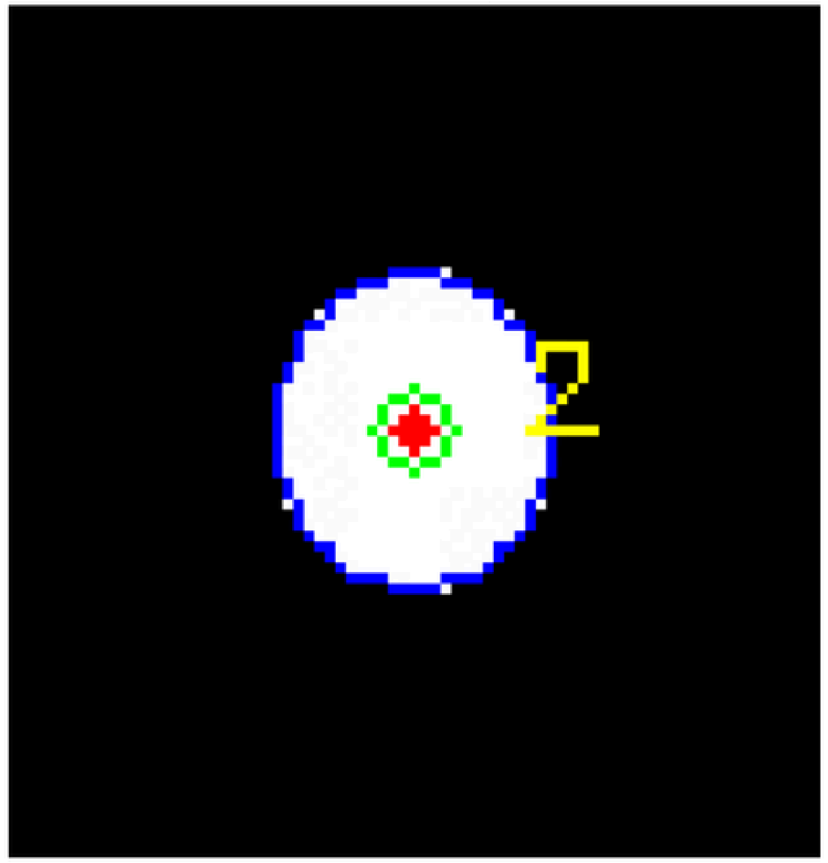
Circular Target Tracking

Bounding boxes, target position, deviation correction, iteratively for consecutive frames

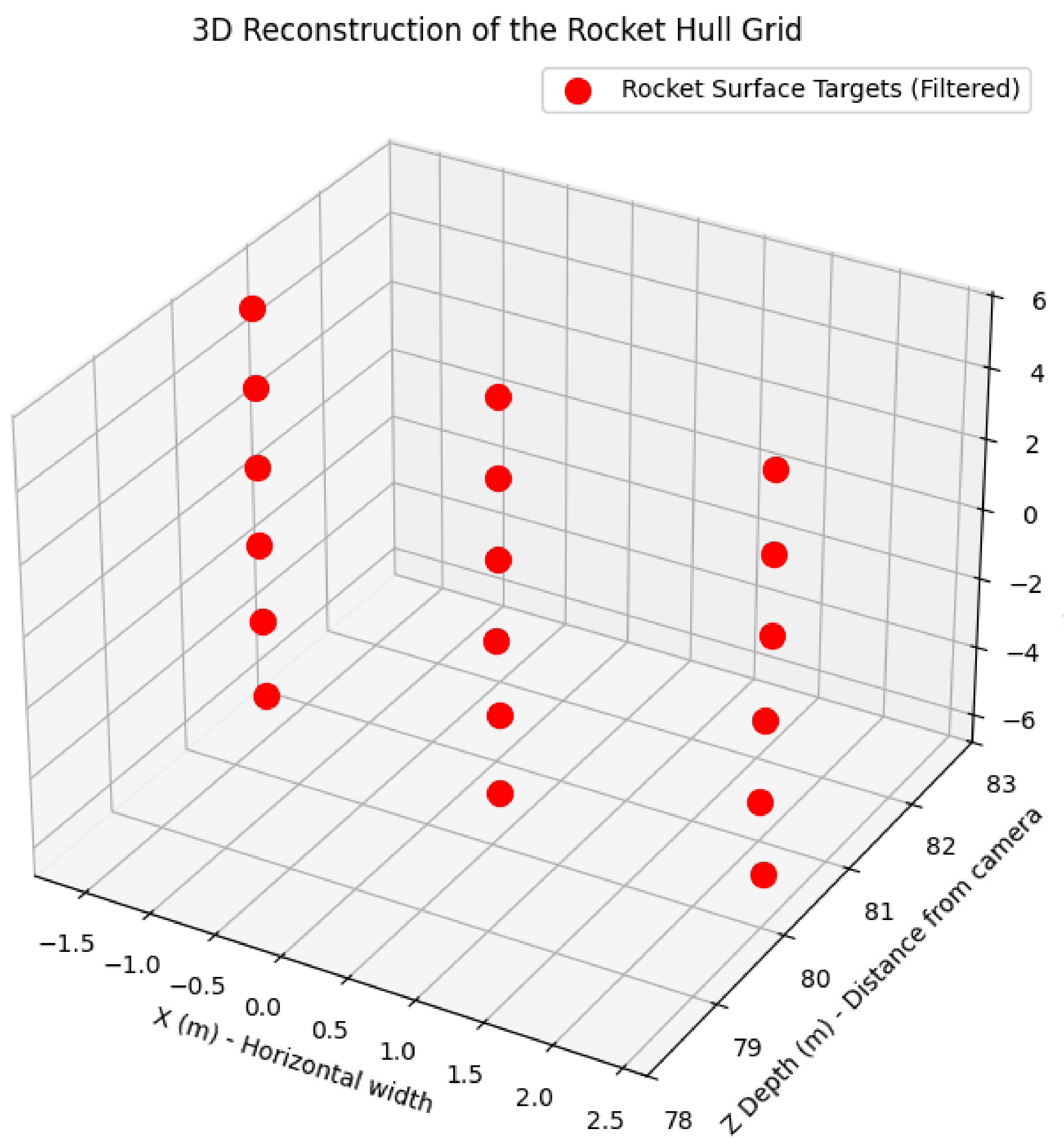
Full View (Hollow Green = Geometric, Solid Red = True Center)

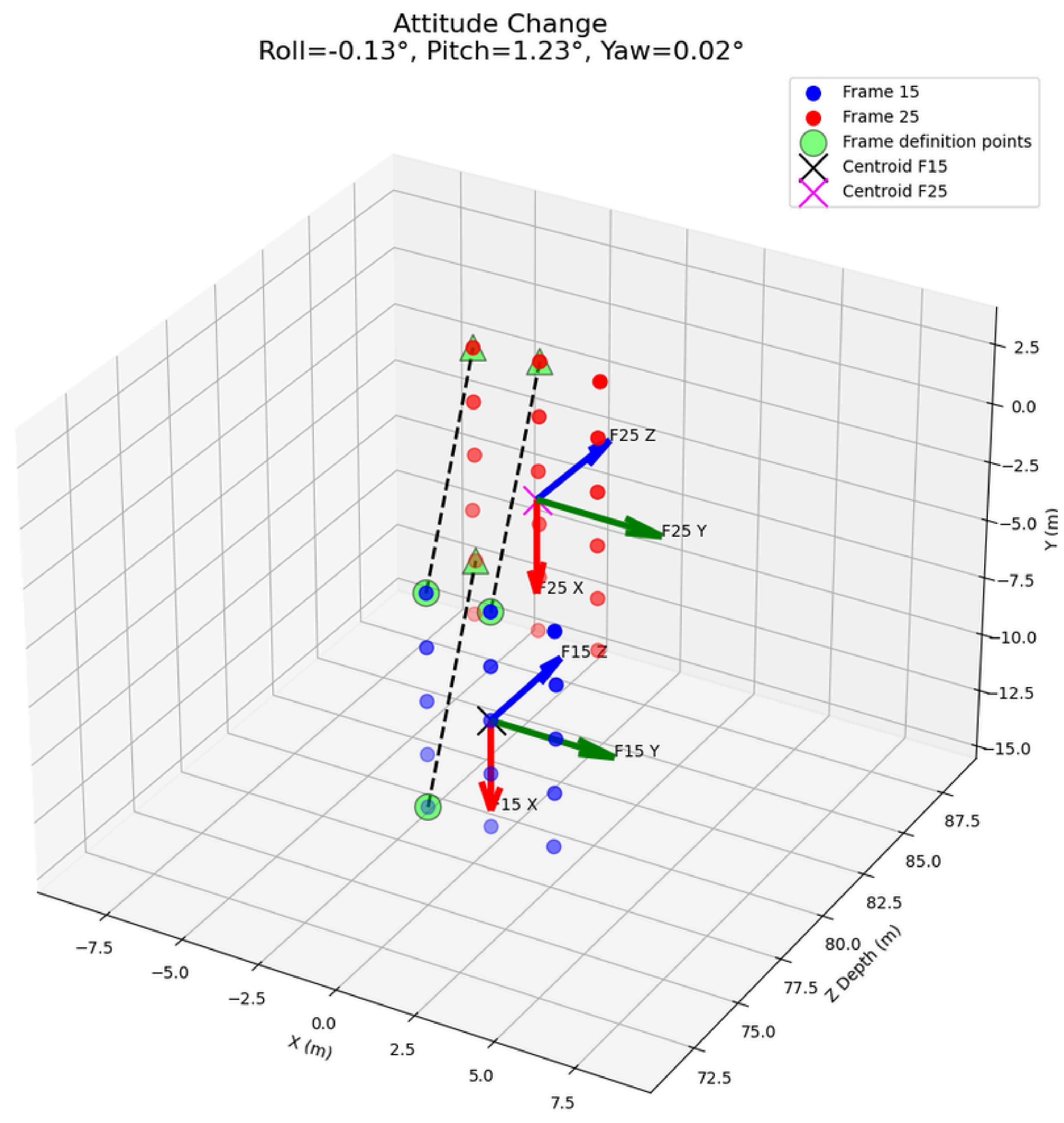


Zoomed-In View of Target 2
(Max Shift: 0.040 pixels)

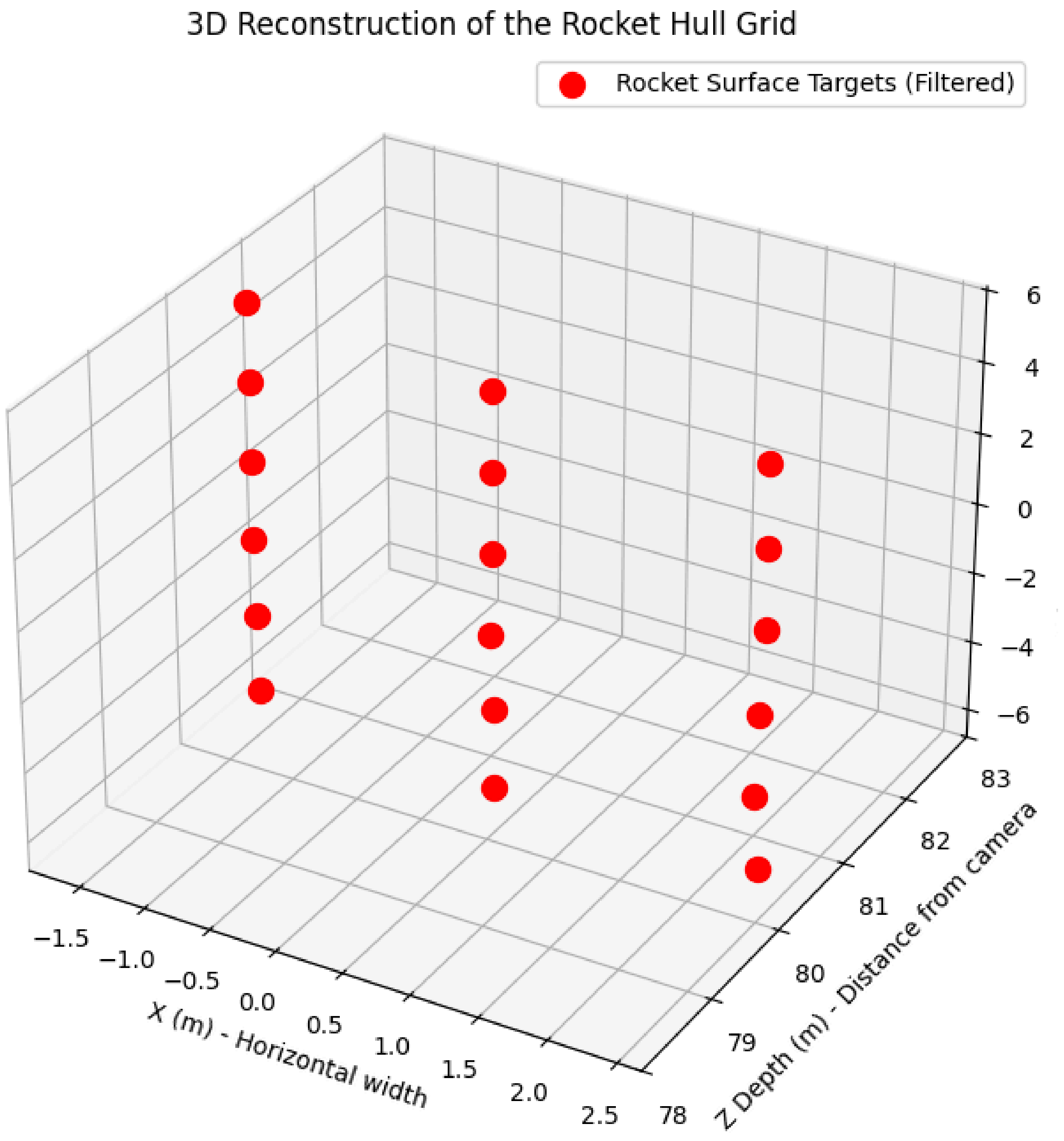


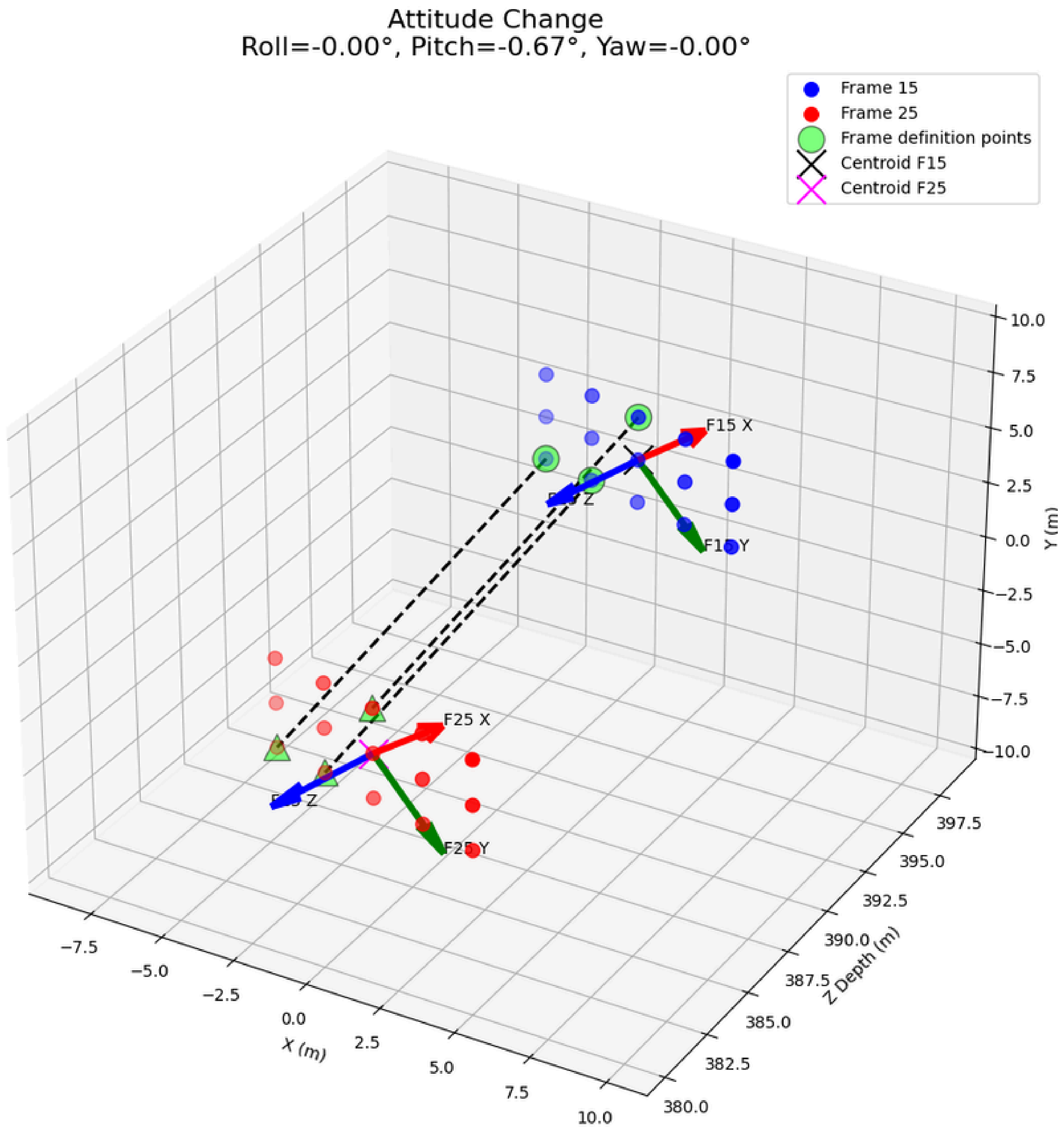
Triangulation and Attitude using Singular Value Decomposition(SVD)





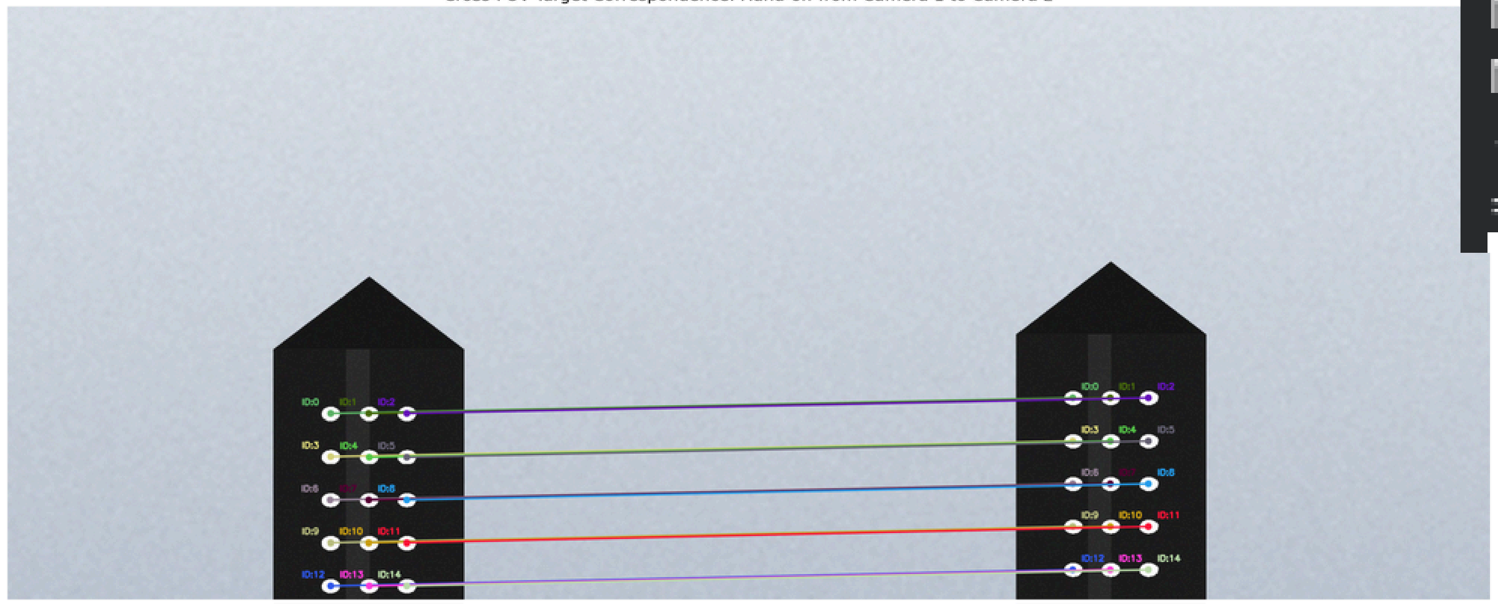
RLV Pose Estimation during Horizontal Landing





Circular Target Matching

Cross-FOV Target Correspondence: Hand-off from Camera 1 to Camera 2



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 PIPELINE ACCURACY EVALUATION (NOISE LEVEL: 40.0)
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Total Targets Tracked        : 15
Mean Positioning Error (RMSE): 69.4599 mm
Maximum Point Deviation      : 136.2679 mm
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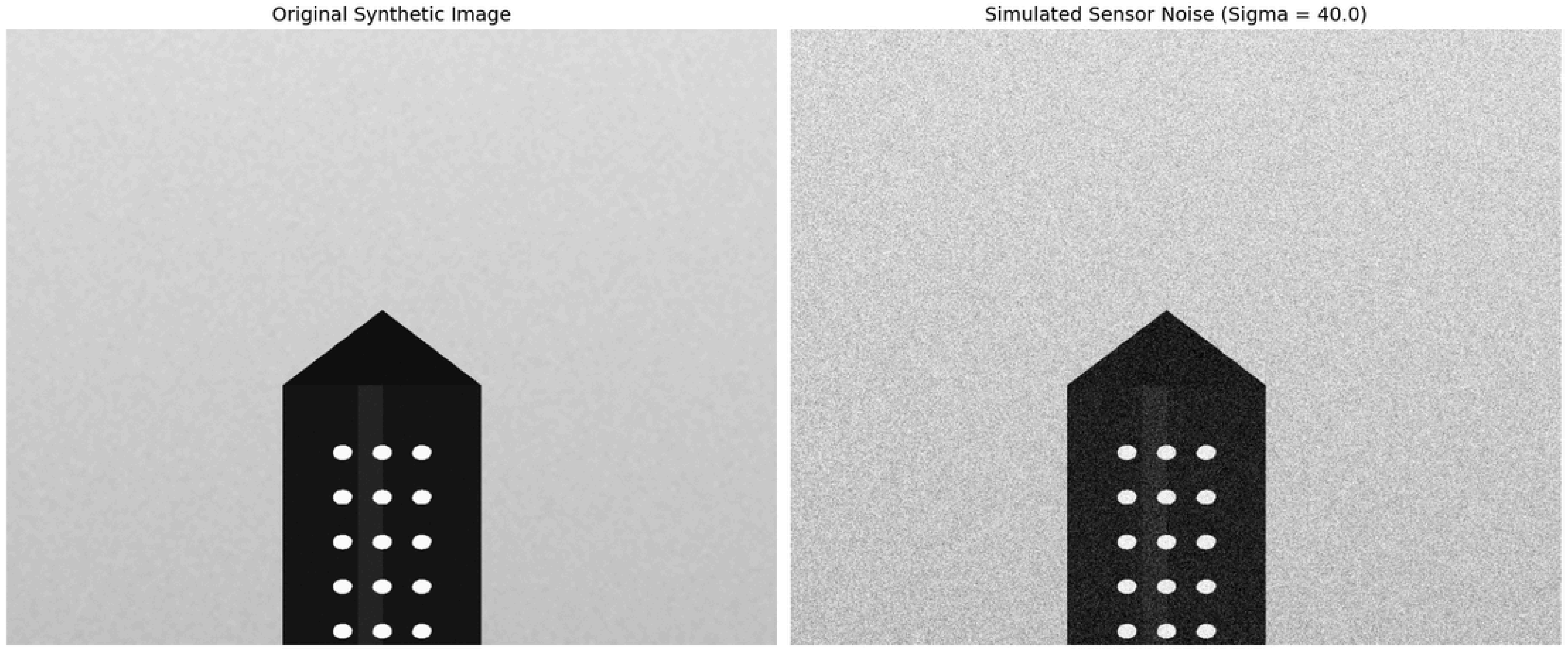

Cameras arrangement and Parameters

Parameter	Recommended
Sensor	Global shutter CMOS, 1280×1024–2048×1536
Frame rate	200–500 fps
Exposure	≤ 200 μs (electronic shutter)
Sync	Hardware trigger, μs-level jitter across all gates
Lens (touchdown zone gates)	~30–35 mm
Lens (rollout zone gates)	~16–25 mm
Baseline per gate	30–50 m (2× standoff)
Standoff per gate	15–25 m from centerline
Toe-in/convergence	Aimed at expected flight-path altitude for that gate's phase
Along-track coverage/gate	10–25 m depending on lens
Gate spacing, touchdown zone	~8–20 m (dense, overlapped)
Gate spacing, rollout zone	50–100 m (sparse)
Occlusion mitigation	Add 3rd narrow-baseline camera per gate (Xu et al. trinocular concept) for crab-angle/bank occlusion recovery
Targets	Circular retro-reflective/white targets (Tong et al.) if surface allows, or CNN keypoint tracking on natural features (nose, wingtips, gear) if surface is tiled/textured like a real thermal-protection skin